



ZIMBABWE INTEGRATING CAPITAL AND RISK PROGRAMME

('ZICARP')

TS 6.3 – Determination of Non-Life Underwriting Risk Capital



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1. Introduction

- 1.1 Underwriting in non-life insurance is the process of evaluating the risk of insuring for example a house, car driver or properties etc, to determine if it's profitable for the insurance company to take the chance of providing insurance.
- 1.2 Non-life underwriting risk is the risk arising from non-life insurance obligations, such as poor claims experience, expense over-runs and policy lapses. Underwriting risk in general refers to the loss, or adverse change in the value of insurance liabilities, due to inadequate pricing and provisioning.
- 1.3 The Solvency Capital Requirement ('SCR') charge in respect of non-life underwriting risk covers the risk of loss arising from non-life insurance obligations.
- 1.4 There are three sources of non-life risk covered under the Zimbabwe Integrating Capital and Risk Programme ('ZICARP'): premium risk, reserve risk and catastrophe risk.

Objectives

- 1.5 The calculation of the capital requirement for Non-Life Underwriting Risk ('NLUR') is based on a mixture of linear formulae and specified scenarios applied to the following risk components of non-life underwriting risk:
- 1.5.1 Premium risk;
 - 1.5.2 Reserve risk; and
 - 1.5.3 Catastrophe risk.
- 1.6 The overall capital requirement for non-life underwriting risk is determined by aggregating the capital requirements of premium, reserve and catastrophic risk including the correlation matrix.

2. NLUR Solvency Capital Requirement

2.1 The NLUR SCR is derived by combining the SCRs for the three non-life underwriting risk sub-modules using a correlation matrix.

2.2 The SCR for each sub-module is taken as the impact on the claims and the premium reserves and net earned premium.

2.3 The capital requirement for non-life underwriting risk (SCR_{NL}) must be calculated by combining the capital requirements for each non-life underwriting risk component using the following formula:

$$SCR_{NL} = \sqrt{\sum_{r,c} CorrNL_{r,c} \cdot NL_r \cdot NL_c}$$

where: $CorrNL_{r,c}$ are the entries of the correlation matrix $CorrNL$; and $NL_r \cdot NL_c$ are the capital requirements for individual non-life underwriting risk components r and c according to the rows and columns of correlation matrix $CorrNL$.

3. Estimation of NLUR Sub-Risk Capital Requirements

Premium Risk

- 3.1 Non-life premium risks refer to the risk of loss, or adverse change in the value of insurance liabilities, resulting from fluctuations in the timing, frequency and severity of insured events and in the time of claim settlements.
- 3.2 Premium risk results from fluctuations in the timing, frequency and severity of insured events. For primary insurance business, premium risk relates to policies to be written (including renewals) during the period, and to unexpired risks on existing contracts.
- 3.3 Premium risk includes the risk that premium provisions turn out to be insufficient to cover claims or need to be increased.
- 3.4 The capital requirement for premium risk shall be computed as the sum of the product of risk factors and the premiums reserves for sub classes of insurance business.
- 3.5 The premium reserve is the sum of the unearned premium and additional unexpired risk.
- 3.6 The risk factors considered for premium risk are as given in section 6 of this technical specification.

Reserve Risk

- 3.7 Reserve risk results from fluctuations in the timing and amount of claim settlements.
- 3.8 The capital requirement for reserve risk shall be computed as the sum of the product of risk factors and the claims reserves for sub classes of insurance business.
- 3.9 The claim reserve is the sum of incurred but not reported (IBNR), Outstanding Claims (OCR) and incurred but not enough reported (IBNER).
- 3.10 The risk factors considered for reserve risk are as given in section 6 of this technical specification.

Catastrophe Risk

- 3.11 Catastrophe (CAT) risk is the risk of loss or of adverse change in the value of insurance liabilities, resulting from significant uncertainty of pricing and provisioning assumptions related to extreme or exceptional events.
- 3.12 CAT risks stem from extreme or irregular events that are not sufficiently captured by the capital requirements for premium and reserve risk.
- 3.13 The capital required for catastrophic risk shall be computed as the sum of the product of risk factors and the net earned premiums for each class of business.
- 3.14 The risk factors considered for catastrophe risk are as given in section 6 of this technical specification.

4. Determination of correlation

Correlation Matrix for Premium and Reserve Risk Calculation

4.1 Correlation matrix *CorrNL* is defined as:

Table 1: Correlation matrix *CorrNL*

<i>CorrNL</i>	Premium Risk	Reserve Risk	Catastrophic Risk
Premium Risk	100%	25%	0%
Reserve Risk	25%	100%	0%
Catastrophic Risk	0%	0%	100%

4.2 *CorrNL* is the correlation of subcomponents of non-life underwriting. It is assumed that the correlation between catastrophe risk and the other risk components is zero since the subcomponents are independent.



5. Conclusion

- 5.1 The risk factors for non-life underwriting risk capital requirement are subject to continuous review and updating taking into account developments in the macro-economic environment.

6. TS6.3 A1 – NLUR Capital Charges

The table below shows the average of premiums, claims and catastrophe capital charges¹.

Main Class	Sub-class 1	Premium Capital Charge	Claims Capital Charge	Catastrophic Capital Charge
	Liability	25%	27%	5%
	Engineering all risks	32%	26%	5%
	Fire	30%	26%	5%
	Farming	30%	26%	5%
	Motor	22%	19%	5%
	Aviation	34%	30%	5%
	Marine Bonds	29%	23%	5%
		30%	32%	5%
Financial Loss	Hire Purchase	30%	32%	5%
Fixed Benefits	Accident Travel Insurance	23%	20%	5%
		34%	30%	5%
	Legal	25%	27%	5%
	Liability	36%	22%	5%
	Fidelity Guarantee	30%	32%	5%

¹ Factors are based on the average East Africa experience